

DECENTRALIZED NOTARY SMART CONTRACT

Remix IDE – Test Evidence

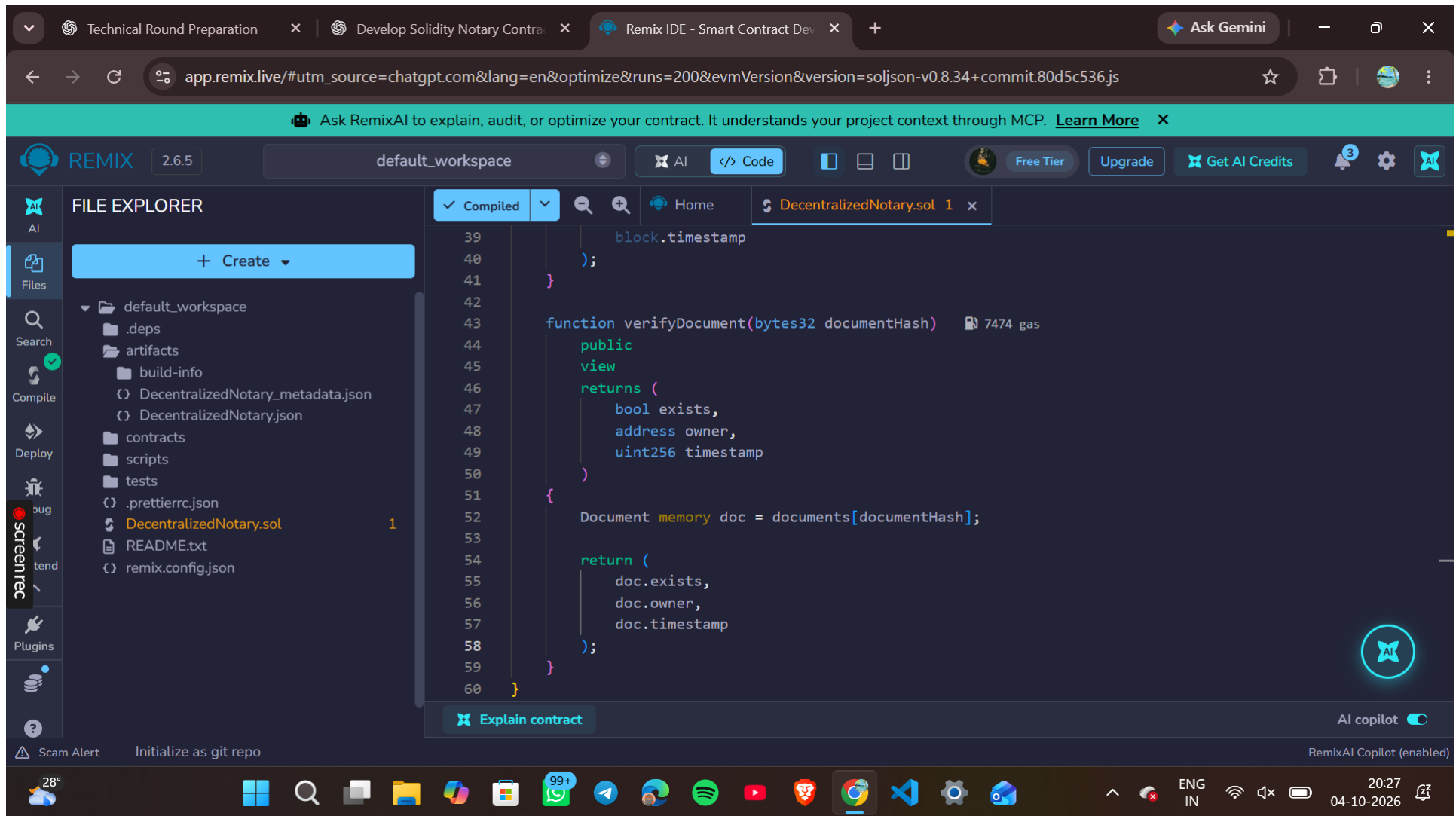
This document contains the implementation and testing evidence for the Decentralized Notary Smart Contract developed using Solidity and Remix IDE.

Project	Decentralized Notary Smart Contract
Language	Solidity ^0.8.20
Development Tool	Remix IDE
Execution Environment	Remix VM (Cancun)
Core Functions	registerDocument(), verifyDocument()

Prepared for academic project submission

1. Solidity Source Code

DecentralizedNotary.sol source code shown in Remix IDE.

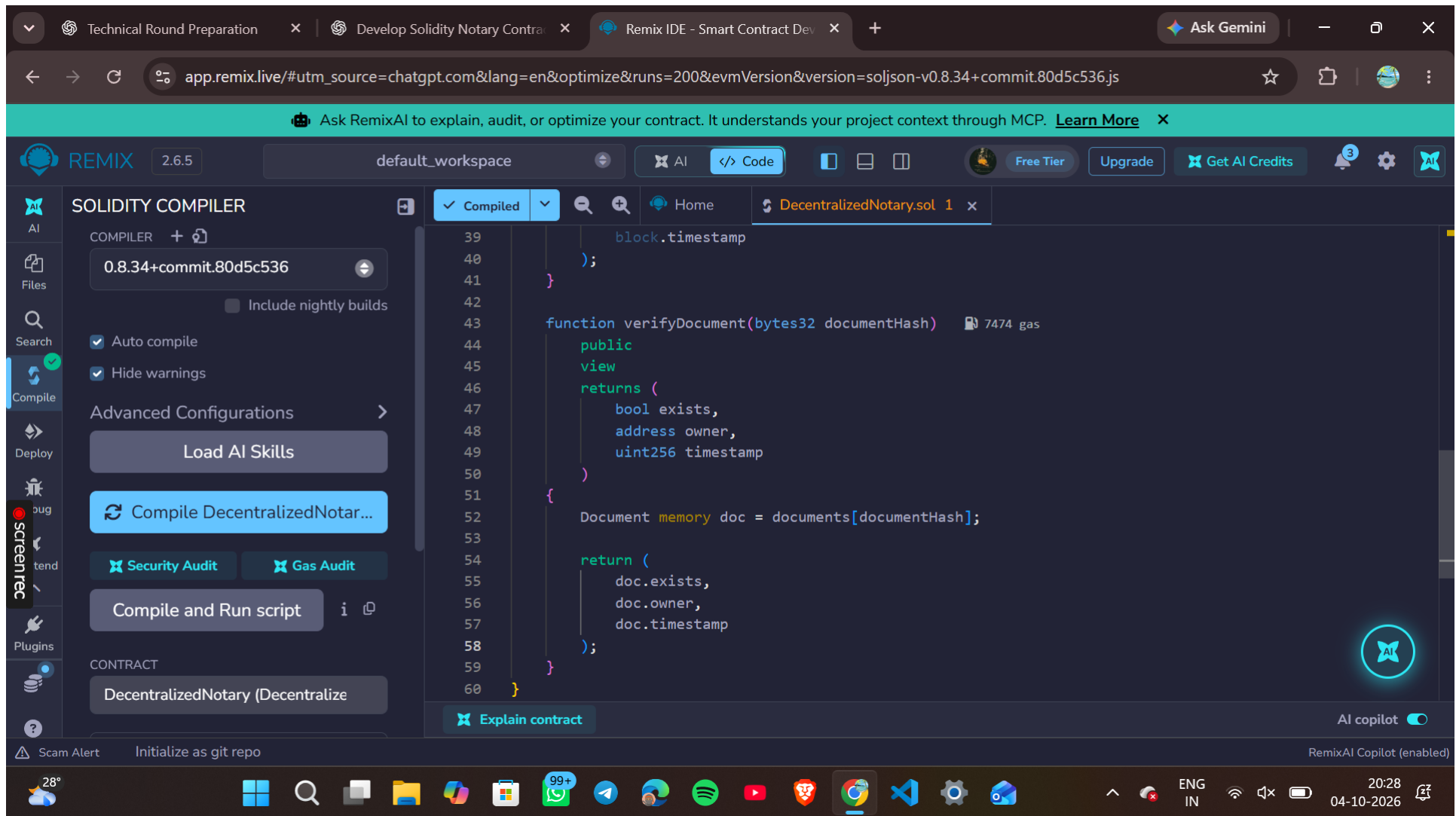


The screenshot displays the Remix IDE interface. The top navigation bar shows the current workspace as 'default_workspace' and the active file as 'DecentralizedNotary.sol'. The left sidebar contains the 'FILE EXPLORER' with a search bar and a list of files and folders, including 'default_workspace', '.deps', 'artifacts', 'build-info', 'DecentralizedNotary_metadata.json', 'DecentralizedNotary.json', 'contracts', 'scripts', 'tests', '.prettierrc.json', 'DecentralizedNotary.sol', 'README.txt', and 'remix.config.json'. The central code editor shows the Solidity source code for 'DecentralizedNotary.sol'. The code includes a 'block.timestamp' reference, a 'verifyDocument' function, and a 'return' statement. The bottom status bar indicates 'Scam Alert' and 'Initialize as git repo'. The system tray at the bottom shows the date and time as '04-10-2026 20:27'.

```
39         block.timestamp
40     };
41 }
42
43 function verifyDocument(bytes32 documentHash) 7474 gas
44     public
45     view
46     returns (
47         bool exists,
48         address owner,
49         uint256 timestamp
50     )
51 {
52     Document memory doc = documents[documentHash];
53
54     return (
55         doc.exists,
56         doc.owner,
57         doc.timestamp
58     );
59 }
60 }
```

2. Contract Compilation

The DecentralizedNotary contract is successfully compiled using Solidity 0.8.34.



3. Contract Deployment

DecentralizedNotary is successfully deployed in Remix VM (Cancun).

The screenshot displays the Remix IDE interface with the following components:

- Top Bar:** Includes browser tabs for "Technical Round Preparation", "Develop Solidity Notary Contra...", and "Remix IDE - Smart Contract Dev...". The address bar shows `app.remix.live/#utm_source=chatgpt.com&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js`. A banner for the "Learneth plugin" is visible.
- Left Sidebar:** Contains navigation icons for AI, Files, Search, Compile, Deploy, Bug, Tendency, and Plugins.
- Environment Panel:** Shows the current environment as "Remix VM" and "Cancun". It displays "Account 1" with address `0x5B3...eddC4` and a balance of "99.999 ETH".
- Deployed Contracts Panel:** Lists "Deployed Contracts" with a count of 1. It shows the "DecentralizedNotary" contract deployed to the "Remix VM" at address `0xd91...39138` "2s ago".
- Code Editor:** Displays the Solidity code for `DecentralizedNotary.sol`. The visible code includes:

```
emit DocumentRegistered(  
    documentHash,  
    msg.sender,  
    block.timestamp  
);
```
- Console/Logs Panel:** Shows the output of the deployment transaction:

```
[vm] from: 0x5B3...eddC4 to: DecentralizedNotary.(constructor) value: 0 wei data: 0x608...20033 logs: 0  
hash: 0xbcf...69c07
```
- Bottom Bar:** Includes a "Scam Alert" notification, a status bar with "RemixAI Copilot (enabled)", and a Windows taskbar at the very bottom showing the time as 20:31 on 04-10-2026.

4. Deployed Contract and Functions

The deployed contract exposes `registerDocument()` and `verifyDocument()`.

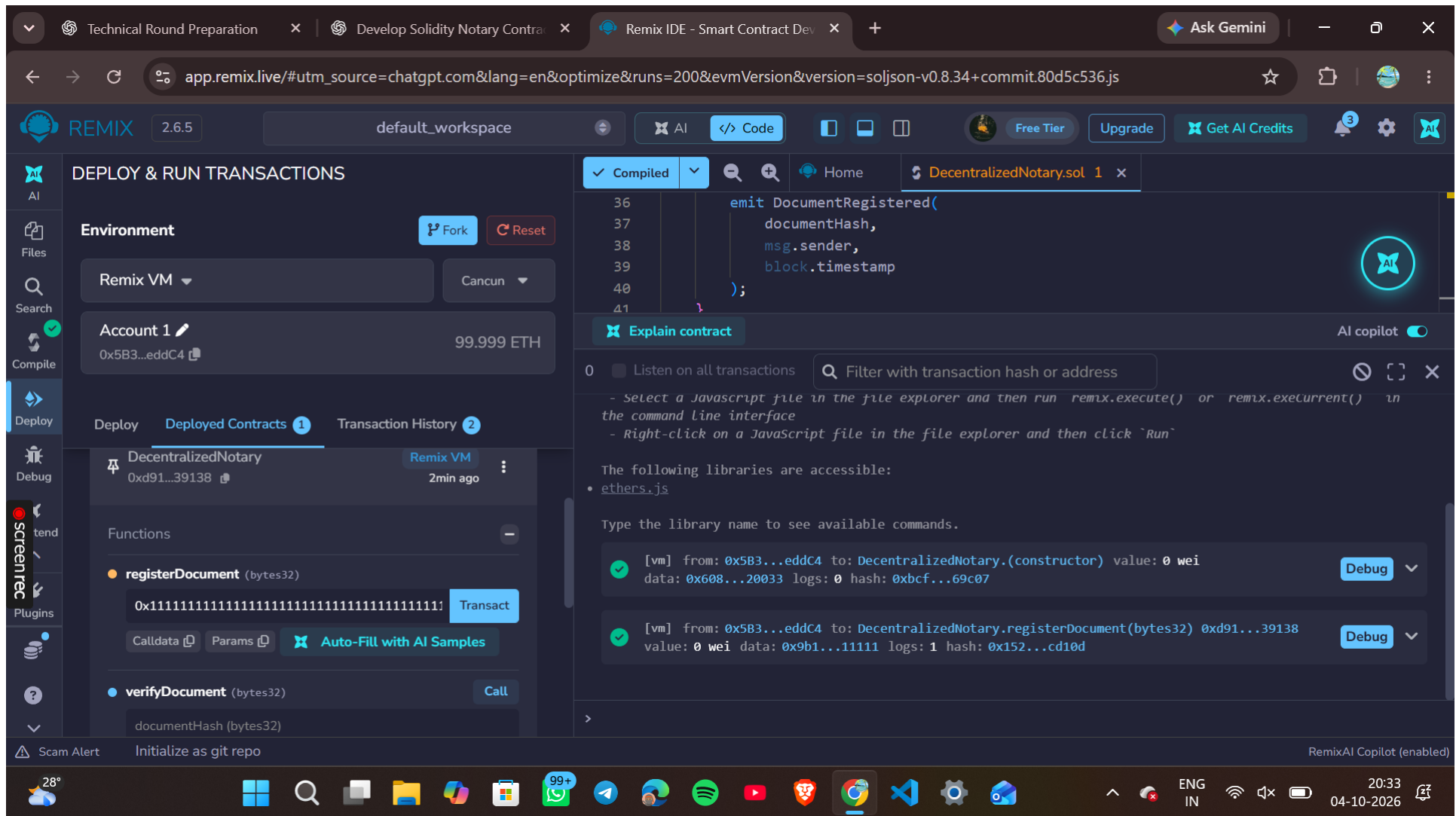
The screenshot displays the Remix IDE interface, which is used for developing and deploying smart contracts. The interface is divided into several panels:

- Top Panel:** Shows the browser address bar with the URL `app.remix.live/#utm_source=chatgpt.com&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js`. The Remix logo and version (2.6.5) are visible.
- Left Panel:** Contains navigation icons for AI, Files, Search, Compile, Deploy, Debug, and Plugins. The **Deploy** section is active, showing the **Deployed Contracts** tab with a list of deployed contracts. One contract, **DecentralizedNotary**, is listed with its address `0xd91...39138` and a button to **Transact**.
- Center Panel:** Displays the **Code** editor with the `DecentralizedNotary.sol` file. The code shows a function `emit DocumentRegistered` that takes `documentHash`, `msg.sender`, and `block.timestamp` as arguments. Below the code editor, there is an **Explain contract** section with instructions on how to use the contract and a list of accessible libraries (e.g., `ethers.js`).
- Right Panel:** Shows the **Transaction History** tab, which displays a list of transactions. A transaction is shown with its status (green checkmark), from address (`0x5B3...eddC4`), to address (`DecentralizedNotary.constructor`), value (`0 wei`), data (`0x608...20033`), logs (`0`), and hash (`0xbcf...69c07`).

The bottom of the image shows the Windows taskbar with various application icons and the system clock indicating the time is 20:32 on 04-10-2026.

5. Document Registration – Success

A new document hash is registered successfully.



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6. Document Verification – Success

The registered document returns exists = true, owner address, and timestamp.

The screenshot displays the Remix IDE interface with the following components:

- Top Bar:** Includes browser tabs for 'Technical Round Preparation', 'Develop Solidity Notary Contract', and 'Remix IDE - Smart Contract Dev'. The address bar shows `app.remix.live/#utm_source=chatgpt.com&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js`. The Remix logo and version '2.6.5' are visible, along with buttons for 'AI', 'Code', 'Home', 'Free Tier', 'Upgrade', and 'Get AI Credits'.
- Left Panel:** Contains navigation icons for AI, Files, Search, Compile, Deploy, Debug, and Plugins. The 'Deploy' section is active, showing 'Deployed Contracts' with a list of contracts.
- Deployed Contracts:** A table listing deployed contracts. The first entry is 'verifyDocument (bytes32)' with a 'Call' button. Below the table, the 'Low level interaction' section shows the following data:
 - 0:bool: exists true
 - 1:address: owner 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4
 - 2:uint256: timestamp 1791126201
- Code Editor:** Displays the Solidity code for 'DecentralizedNotary.sol'. The code includes an `emit DocumentRegistered` event with parameters `documentHash`, `msg.sender`, and `block.timestamp`.
- Right Panel:** Features an 'Explain contract' section with an 'AI copilot' toggle. Below it, a list of accessible libraries includes `ethers.js`. The bottom section shows transaction logs with details for VM and CALL operations, including from/to addresses, values, data, logs, and hashes.
- Bottom Bar:** Includes a 'Scam Alert' icon, a button to 'Initialize as git repo', and the 'RemixAI Copilot (enabled)' status.

7. Duplicate Registration – Rejected

A second attempt with the same hash is rejected with "Document already registered".

The screenshot displays the Remix IDE interface with the following components:

- Top Bar:** Includes browser tabs for "Technical Round Preparation", "Develop Solidity Notary Contract", and "Remix IDE - Smart Contract Dev". The address bar shows `app.remix.live/#utm_source=chatgpt.com&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js`. The Remix logo and version "2.6.5" are on the left, and "AI", "Code", and "Free Tier" buttons are on the right.
- Left Panel:** Contains a sidebar with icons for AI, Files, Search, Compile, Deploy, and Debug. The "Deploy" tab is active, showing "Deployed Contracts" with one entry: `0xd91...39138` (3min ago). Below this, the "Functions" section lists `registerDocument (bytes32)` and `verifyDocument (bytes32)`, each with a "Transact" or "Call" button.
- Center Panel:** Displays the Solidity code for `DecentralizedNotary.sol`. Lines 36-41 show the `emit DocumentRegistered` event. Below the code, the "Explain contract" section is visible.
- Right Panel:** Shows a transaction log with a red error icon. The message reads: `[vm] from: 0x5B3...eddC4 to: DecentralizedNotary.registerDocument(bytes32) 0xd91...39138 value: 0 wei data: 0x9b1...11111 logs: 0 hash: 0x071...956bf`. Below this, the error details are shown: `transact to DecentralizedNotary.registerDocument errored: Error occurred: Error occurred: revert.` and `revert Reason provided by the contract: "Document already registered".`
- Bottom Bar:** Includes a "Scam Alert" icon, a status bar with "Initialize as git repo", and system icons for temperature (28°), time (20:35), and date (04-10-2026).

8. Unregistered Document Verification

A different hash returns exists = false, zero address, and timestamp 0.

The screenshot displays the Remix IDE interface. The top bar shows the browser address: `app.remix.live/#utm_source=chatgpt.com&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js`. The left sidebar contains navigation icons for AI, Files, Search, Compile, Deploy, Debug, and Plugins. The main workspace is divided into three panels:

- Environment Panel:** Shows the account `Account 1` with address `0x5B3...eddC4` and a balance of `99.999 ETH`. It also shows the `Deployed Contracts` tab with a contract named `verifyDocument (bytes32)`.
- Code Editor:** Displays the `DecentralizedNotary.sol` file. The code includes a function `emit DocumentRegistered` that takes `documentHash`, `msg.sender`, and `block.timestamp` as arguments.
- Transaction History Panel:** Shows a failed transaction for `verifyDocument`. The transaction data is as follows:
 - `0:bool: exists false`
 - `1:address: owner 0x00000000000000000000000000000000`
 - `2:uint256: timestamp 0`

The bottom status bar indicates a `Scam Alert` and provides an option to `Initialize as git repo`. The system tray at the bottom shows the date and time as `04-10-2026 20:35`.

9. Transaction History

Remix transaction history shows deployment, successful registration, and the expected failed duplicate attempt.

The screenshot displays the Remix IDE interface. The left sidebar contains navigation icons for AI, Files, Search, Compile, Deploy, Debug, Frontend, and Plugins. The main workspace is divided into three panels:

- Environment Panel:** Shows the current environment as 'Remix VM' and 'Account 1' with a balance of 99.999 ETH.
- Transaction History Panel:** Displays a list of transactions:
 - registerDocument** (tx: 0x071...956bf) - **transact** - 2min ago - **Failed** (red error icon).
 - registerDocument** (tx: 0x152...cd10d) - **transact** - 3min ago - **Successful** (green checkmark icon).
 - DecentralizedNotary** (tx: 0xbcf...69c07) - **deploy** - 6min ago - **Successful** (green checkmark icon).
- Code Editor:** Shows the Solidity code for `DecentralizedNotary.sol`. The current function is `emit DocumentRegistered`, which takes `documentHash`, `msg.sender`, and `block.timestamp` as arguments.
- Console:** Shows the execution of a `verifyDocument` function call. The message indicates a `revert` error with the reason: "Document already registered".

The bottom status bar shows the system tray with the date 04-10-2026 and time 20:37.

10. Deployment Transaction Details

Transaction details show the mined status, transaction hash, block hash, and block number.

The screenshot displays the Remix IDE interface with the 'Transaction History' tab selected. The left sidebar contains navigation icons for AI, Files, Search, Compile, Deploy, Debug, Frontend, and Plugins. The main area is divided into three sections: 'Environment' (showing 'Remix VM' and 'Account 1' with 99.999 ETH), 'Deploy' (with tabs for Deploy, Deployed Contracts, and Transaction History), and 'Transaction History' (showing a list of transactions). The 'Transaction History' section lists three transactions: 'registerDocument' (tx: 0x071...956bf, 3min ago), 'registerDocument' (tx: 0x152...cd10d, 4min ago), and 'DecentralizedNotary' (tx: 0xbcf...69c07, 7min ago). The 'DecentralizedNotary' transaction is highlighted. The right sidebar shows the 'DecentralizedNotary.sol' file with the following code:

```
36  
37  
38  
39  
40  
41  
emit DocumentRegistered(  
    documentHash,  
    msg.sender,  
    block.timestamp  
);
```

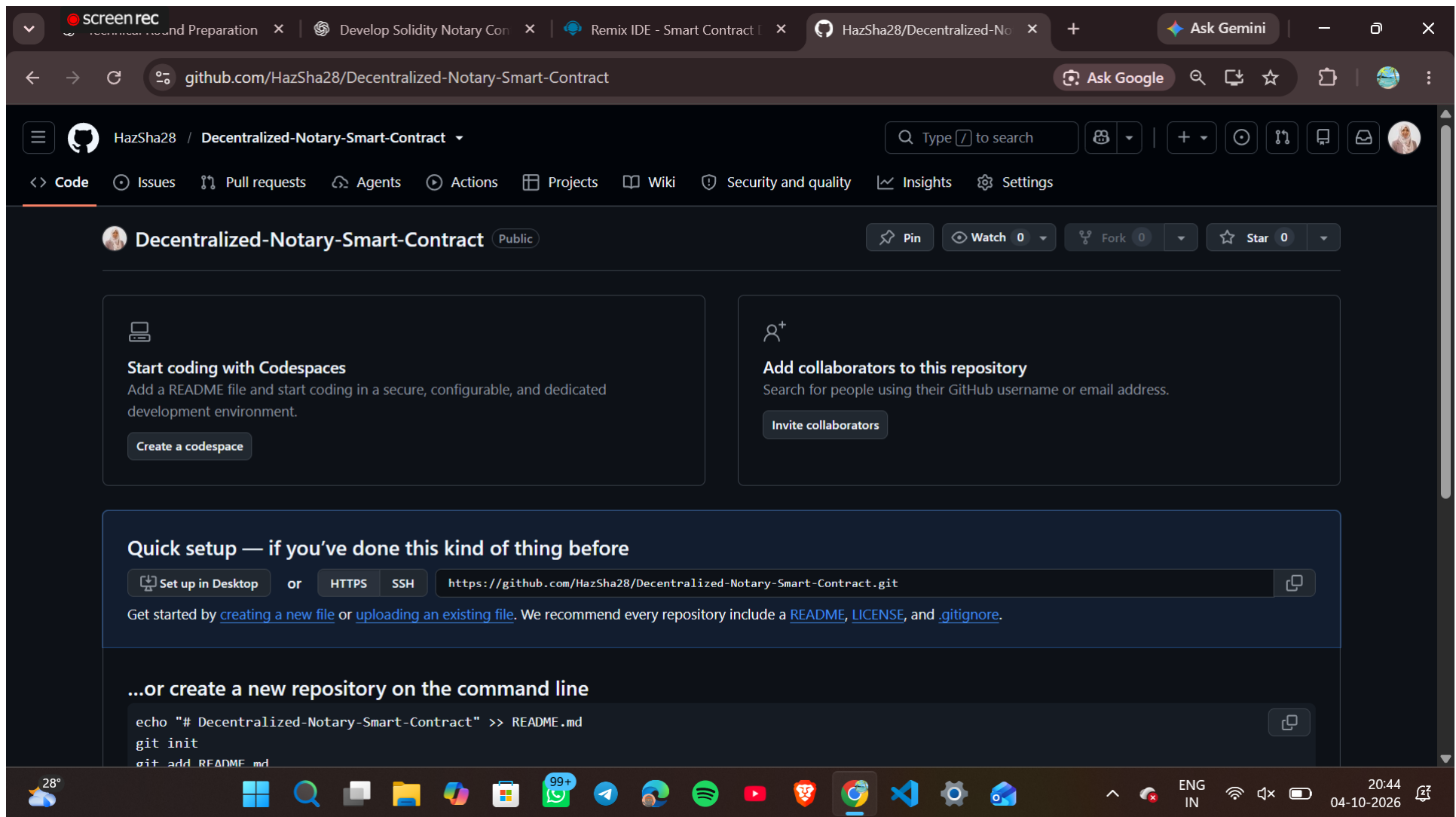
Below the code editor, the 'Explain contract' section is visible, showing the command 'Type the library name to see available commands.' and the output: '[vm] from: 0x5B3...eddC4 to: DecentralizedNotary.(constructor) value: 0 wei data: 0x608...20033 logs: 0 hash: 0xbcf...69c07'. The 'Debug' button is also present. The bottom status bar shows 'Scam Alert', 'Initialize as git repo', and 'RemixAI Copilot (enabled)'.

Transaction	TX Hash	Status	Time Ago
registerDocument	tx: 0x071...956bf	transact	3min ago
registerDocument	tx: 0x152...cd10d	transact	4min ago
DecentralizedNotary	tx: 0xbcf...69c07	deploy	7min ago

Property	Value
status	1 Transaction mined and execution completed
transaction hash	0xbcf913d3fc42b5810ace0bdc385abb8c26a9d2cdae6a1c36fa03c795a8769c07
block hash	0x89ad0e6e126478996a776e88f73f58ef4176a8b45c31ff0b21fa741ef8f0bd1c
block number	1

11. GitHub Repository

The public GitHub repository for the Decentralized Notary Smart Contract.



Test Case Summary

Test Case	Expected Result	Result
Register new document	Document registered successfully	PASS
Verify registered document	exists = true; owner and timestamp returned	PASS
Register duplicate document	Transaction rejected: "Document already registered"	PASS
Verify unregistered document	exists = false; zero address; timestamp = 0	PASS

The screenshots demonstrate successful compilation, deployment, document registration, verification, duplicate protection, and verification of an unregistered document.